

Pediatric Health Guide

Comprehensive Reference for Parents and Caregivers

Covering: Fever Management, Newborn Care, Sleep Safety, Breastfeeding, Developmental Milestones, Emergency Guidance, and Nutrition by Age

Based on guidelines from the American Academy of Pediatrics (AAP), Centers for Disease Control and Prevention (CDC), and World Health Organization (WHO). This document is for educational and informational purposes only and is not a substitute for professional medical advice.

1. Fever in Children: Complete Guide

What Is a Fever?

A fever is a body temperature higher than normal. It is not a disease itself but a sign that the body's immune system is actively fighting an infection. Most fevers are caused by common viral infections and are harmless. In fact, fever is considered a protective mechanism that helps the body fight illness more effectively.

A child has a fever when their temperature reaches or exceeds these thresholds:

- Rectal (most accurate for infants): 100.4°F (38.0°C) or higher
- Oral (for children who can hold a thermometer): 100°F (37.8°C) or higher
- Axillary (armpit, least accurate): 99°F (37.2°C) or higher
- Temporal/forehead: 100.4°F (38.0°C) or higher

Fever Ranges and What They Mean

- **Low-grade fever (100–102°F / 37.8–39°C):** Generally beneficial. The body is fighting infection. Often does not require treatment unless the child is uncomfortable.
- **Moderate fever (102–104°F / 39–40°C):** More significant. Monitor the child closely. Treat for comfort if the child seems uncomfortable, is not drinking, or cannot sleep.
- **High fever (above 104°F / 40°C):** Requires attention. Contact your pediatrician. While high fevers are not automatically dangerous, they warrant evaluation.

Home Treatment for Fever

The goal of treating a fever is to make the child more comfortable, not to bring the temperature to normal. A child with a fever who is playing, drinking, and generally acting normally may not need any treatment at all.

- **Keep the child hydrated:** Offer frequent small sips of water, Pedialyte (oral rehydration solution), breast milk, or formula. Dehydration is the real risk with fever, not the temperature itself. Popsicles count as fluids for older children.
- **Dress lightly:** Remove extra layers of clothing. Dress the child in one light layer. Do not bundle a feverish child in blankets; this traps heat and can raise the temperature further.
- **Room temperature:** Keep the room comfortably cool. A fan on low is acceptable, but do not direct it at the child.
- **Lukewarm bath:** A lukewarm (not cold) sponge bath can help bring the temperature down by 1–2 degrees. Stop if the child starts shivering, as shivering generates more heat. Never use cold water, ice baths, or rubbing alcohol.
- **Acetaminophen (Tylenol):** Can be given to children 2 months and older. Follow the dosing instructions on the package based on the child's weight, or contact your pediatrician or pharmacist for exact dosing. Can be given every 4–6 hours.
- **Ibuprofen (Motrin/Advil):** Can be given to children 6 months and older. Also dosed by weight. Can be given every 6–8 hours. Do not give to children with vomiting, dehydration, or certain kidney conditions.

ALERT: NEVER give aspirin to children or teenagers. Aspirin use in children with viral illness is associated with Reye's syndrome, a rare but potentially fatal condition affecting the brain and liver.

What NOT to Do

- Do not use cold baths or ice baths (causes shivering, which raises body temperature)
- Do not use rubbing alcohol on the skin (can be absorbed and cause poisoning)
- Do not alternate acetaminophen and ibuprofen unless specifically directed by your pediatrician (risk of dosing errors)
- Do not wake a sleeping child to give fever medicine (sleep is healing)
- Do not force food (decreased appetite during fever is normal; focus on fluids)

When to Call the Pediatrician

ALERT: Call your doctor RIGHT AWAY if your child has a fever AND any of these:

- **Any infant under 3 months (12 weeks) with a temperature of 100.4°F (38°C) or higher** — even a slight fever in a young baby can indicate serious infection. This is always an urgent call, day or night.
- Child 3–24 months with a fever of 102.5°F (39.2°C) or higher, especially if acting ill
- Fever above 104°F (40°C) repeatedly at any age
- Looks very ill, unusually drowsy, or extremely fussy
- Has a stiff neck, severe headache, severe sore throat, severe ear pain, or unexplained rash
- Has difficulty breathing
- Has repeated vomiting or diarrhea
- Shows signs of dehydration (no tears when crying, dry mouth, no wet diapers for 6+ hours, sunken eyes)
- Has immune system problems (sickle cell disease, cancer, is on steroids or chemotherapy)
- Has had a seizure
- Fever persists more than 24 hours in a child under 2 years
- Fever persists more than 3 days (72 hours) in a child 2 years or older
- Child still acts sick after the fever is brought down with medicine

Febrile Seizures

Febrile seizures occur in about 2–5% of children between ages 6 months and 5 years. They are triggered by fever and are usually harmless, lasting less than 5 minutes. They do not cause brain damage or epilepsy. During a febrile seizure: place the child on their side on a flat surface, do not put anything in their mouth, do not try to restrain the child, time the seizure, and call 911 if the seizure lasts more than 5 minutes.

Source: American Academy of Pediatrics (AAP), [HealthyChildren.org](https://www.healthychildren.org) — Fever and Your Child, 2020

2. Newborn Care Basics

The First Days at Home

The newborn period covers the first 28 days of life. During this time, your baby is adjusting to life outside the womb and requires attentive care across several key areas.

Feeding

- **Frequency:** Newborns need to eat every 2–3 hours, or 8–12 times per day. Feed on demand — watch for hunger cues like rooting (turning head toward touch on cheek), lip smacking, putting hands to mouth, and fussing. Crying is a late hunger sign.
- **Breastfed babies:** Nurse 10–20 minutes per breast. Breast milk is the ideal nutrition and provides antibodies that protect against infections.
- **Formula-fed babies:** Start with 1–2 ounces per feeding in the first week, gradually increasing to 2–3 ounces by 2 weeks. Follow preparation instructions exactly.
- **Signs of adequate feeding:** 6 or more wet diapers per day by day 4, steady weight gain after initial weight loss (most babies regain birth weight by 10–14 days), baby seems satisfied after feeding.
- **Vitamin D:** All breastfed and partially breastfed infants should receive 400 IU of vitamin D supplement daily, starting in the first few days of life.

Diapering

- **Frequency:** Expect 6–10 wet diapers and 3–4 stools per day in the first few weeks. Breastfed babies may stool more frequently.
- **Stool appearance:** Newborn stool transitions from black/tarry (meconium) in the first 1–2 days to green transitional stool to yellow, seedy, loose stool (breastfed) or tan/yellow, firmer stool (formula-fed). This is all normal.
- **Diaper rash prevention:** Change diapers promptly, clean gently with warm water and a soft cloth, allow the skin to air-dry, and apply a zinc oxide barrier cream (like Desitin or A+D) at every change.
- **Treating mild diaper rash:** Apply thick layer of zinc oxide cream, keep area dry, give diaper-free time when possible. See a doctor if the rash has blisters, pus-filled bumps, doesn't improve in 3 days, or if the baby has fever.

Umbilical Cord Care

- Keep the cord stump clean and dry. Fold the diaper below the stump to expose it to air.
- The stump typically falls off within 1–3 weeks. Do not pull it off.
- Give sponge baths (not tub baths) until the cord falls off and the area is healed.
- Call your pediatrician if: the area around the base becomes red, swollen, or foul-smelling; there is active bleeding (a few drops is normal when the cord separates); the cord has not fallen off after 3 weeks.

Bathing

- **Sponge baths only** until the umbilical cord stump falls off and heals (and until circumcision heals, if applicable).

- **Tub baths:** 2–3 times per week is sufficient. Daily baths can dry out newborn skin.
- Use warm (not hot) water. Test with your elbow or inner wrist.
- Use mild, fragrance-free baby soap. Wash the face with water only.
- Never leave a baby unattended in water, even for a moment.

Normal Newborn Appearance (Things That Look Concerning But Are Normal)

- **Baby acne:** Small red or white bumps on the face, appearing around 2–4 weeks. Resolves on its own in weeks to months. No treatment needed. Do not apply lotions or acne products.
- **Milia:** Tiny white bumps on the nose, chin, or cheeks. These are blocked skin pores. Disappear on their own within a few weeks.
- **Cradle cap:** Yellowish, scaly, crusty patches on the scalp. Harmless. Can gently brush with a soft brush and wash with baby shampoo. Resolves by 6–12 months.
- **Stork bites / angel kisses:** Pink or red flat birthmarks on the eyelids, forehead, or back of neck. Very common. Usually fade within 1–2 years.
- **Mongolian spots:** Blue-gray flat patches on the lower back or buttocks, common in babies with darker skin tones. Harmless and usually fade by age 5.
- **Crossed eyes:** Occasional crossing is normal in the first 3–4 months as eye muscles develop. Persistent crossing after 4 months should be evaluated.
- **Hiccups:** Very common in newborns, especially after feeding. Harmless. No treatment needed.
- **Sneezing:** Frequent sneezing is normal and helps clear nasal passages. It does not mean the baby has a cold.
- **Spitting up:** Small amounts of milk coming up after feeding is normal (reflux). Projectile vomiting (forceful, shoots across the room) is not normal — call your pediatrician.

When to Call the Doctor for a Newborn

ALERT: Call your pediatrician immediately if your newborn shows any of these signs:

- Rectal temperature of 100.4°F (38°C) or higher (or below 97.7°F / 36.5°C)
- Refusing to feed or feeding poorly for more than 2 consecutive feedings
- Fewer than 6 wet diapers per day after day 4 of life
- Excessive sleepiness or difficulty waking for feedings
- Persistent or forceful (projectile) vomiting
- Yellow discoloration of skin or whites of eyes (jaundice) worsening after day 3
- Difficulty breathing: grunting, flaring nostrils, retractions (ribs visible with each breath), or breathing faster than 60 breaths per minute
- Blue or gray color of lips, tongue, or skin
- Redness, swelling, discharge, or foul smell around the umbilical cord
- Any signs of dehydration: dry mouth, no tears, sunken soft spot (fontanelle)

3. Sleep Safety and SIDS Prevention

Sudden Infant Death Syndrome (SIDS) is the unexplained death of a baby younger than 1 year, usually occurring during sleep. While the exact cause is unknown, research has identified several evidence-based practices that dramatically reduce the risk.

The ABCs of Safe Sleep

A = Alone, B = Back, C = Crib — This is the foundation of safe infant sleep.

Back to Sleep

- **Always place babies on their BACK** for every sleep — naps and nighttime. This single practice has reduced SIDS deaths by over 50% since the 1990s.
- Back sleeping is recommended until age 1, even after the baby can roll over on their own. Place them on their back; if they roll to their stomach independently, you do not need to reposition them.
- Side sleeping is NOT safe and is NOT recommended. Babies placed on their sides can easily roll onto their stomachs.

Safe Sleep Surface

- **Use a firm, flat mattress** in a safety-approved crib, bassinet, or play yard. The mattress should be covered with only a fitted sheet.
- **Nothing else in the crib:** No blankets, pillows, bumper pads, stuffed animals, positioners, wedges, or loose bedding. These are suffocation and entrapment hazards.
- The sleep surface should be flat (not inclined). Inclined sleepers, car seats, swings, and bouncers are not safe for unsupervised sleep.
- Do not use sleep positioners or nests (such as DockATot or Snuggle Me). These have not been shown to be safe and the FDA has warned against them.

Room Sharing (Not Bed Sharing)

- **Room sharing is recommended** for at least the first 6 months, ideally the first year. Place the crib or bassinet in the parents' bedroom, within arm's reach.
- **Bed sharing is NOT recommended.** Sharing a bed with an infant increases the risk of SIDS, suffocation, and entrapment. This includes sharing with parents, siblings, or pets.
- The risk of bed sharing is especially high if: a parent smokes, has consumed alcohol or sedating medications, the baby is younger than 4 months, the baby was born preterm or with low birth weight, or the sleep surface is soft (couch, armchair, waterbed).

Additional SIDS Risk Reduction

- **Offer a pacifier at sleep time:** Pacifier use during sleep has been associated with reduced SIDS risk. If breastfeeding, wait until breastfeeding is well established (usually 3–4 weeks) before introducing a pacifier. If it falls out during sleep, do not replace it.
- **Avoid overheating:** Dress the baby in no more than one additional layer than what an adult would wear. Watch for signs of overheating: sweating, damp hair, flushed cheeks, heat rash, rapid breathing. Use a sleep sack instead of loose blankets.

- **Breastfeed if possible:** Breastfeeding is associated with a reduced risk of SIDS. Any amount helps — even partial breastfeeding is protective.
- **Do not smoke during pregnancy or after birth.** Smoking is one of the strongest risk factors for SIDS. Do not allow smoking near the baby or in the home.
- **Prenatal care:** Regular prenatal care and avoiding alcohol and drugs during pregnancy reduces SIDS risk.
- **Supervised tummy time when awake:** Tummy time while the baby is awake and observed helps strengthen neck and shoulder muscles and prevents flat head syndrome (positional plagiocephaly). Start from birth for short periods and increase gradually.
- **Keep immunizations up to date:** Evidence suggests that fully immunized infants have a lower risk of SIDS.

How Much Sleep Do Babies Need?

- Newborns (0–3 months): 14–17 hours per day (in short stretches of 2–4 hours)
- Infants (4–11 months): 12–16 hours per day (including naps)
- Toddlers (1–2 years): 11–14 hours per day (including naps)
- Preschoolers (3–5 years): 10–13 hours per day (may or may not nap)
- School-age (6–12 years): 9–12 hours per day
- Teenagers (13–18 years): 8–10 hours per day

Source: AAP Policy Statement on Safe Sleep, 2022; [HealthyChildren.org](https://www.healthychildren.org) — A Parent's Guide to Safe Sleep

4. Breastfeeding Basics

Breast milk is the ideal nutrition for infants. The AAP recommends exclusive breastfeeding for about the first 6 months of life, with continued breastfeeding alongside complementary foods for 2 years or beyond, as mutually desired by mother and child.

Benefits of Breastfeeding

For the baby:

- Provides ideal nutrition with the perfect balance of proteins, fats, and vitamins
- Contains antibodies that help protect against infections (ear infections, respiratory infections, diarrhea, bacterial meningitis)
- Associated with lower rates of SIDS, childhood obesity, type 1 and type 2 diabetes, asthma, and eczema
- Easy to digest; lower rates of constipation and stomach upset
- Adapts to the baby's changing nutritional needs over time

For the mother:

- Helps the uterus return to pre-pregnancy size more quickly
- Associated with lower rates of breast cancer, ovarian cancer, and type 2 diabetes
- Promotes bonding through skin-to-skin contact
- Convenient (always available, right temperature, no preparation needed)
- Saves money compared to formula

How to Breastfeed: Latch and Positioning

- **Good latch signs:** The baby's mouth covers a large portion of the areola (not just the nipple), the lips are flanged outward (like fish lips), you hear rhythmic sucking and swallowing, and you feel a tugging sensation but not sharp pain.
- **Poor latch signs:** Pain that persists throughout the feeding, clicking sounds, the baby's cheeks dimple inward, the nipple looks pinched or flattened after feeding.
- **Common positions:** Cradle hold (baby across your lap, belly to belly), cross-cradle hold (supporting baby's head with opposite hand), football/clutch hold (baby tucked alongside your body), side-lying (both lying on your sides).
- **Tips for success:** Bring the baby to the breast, not the breast to the baby. Support your breast in a C-hold (thumb on top, fingers underneath). Tickle the baby's lip with your nipple to encourage them to open wide. Aim the nipple toward the roof of the baby's mouth.

Feeding Schedule and Duration

- **Newborns:** Feed on demand, usually every 1.5–3 hours (8–12 times per day). Frequent feeding establishes milk supply.

- **Duration:** Allow the baby to feed until they release on their own, then offer the other breast. Feedings typically last 10–20 minutes per side.
- **Signs the baby is getting enough:** 6+ wet diapers per day by day 4, 3–4 stools per day in the first month, steady weight gain, baby seems satisfied after feeding.
- **Growth spurts:** Babies cluster feed (feed more frequently) during growth spurts, typically around 2–3 weeks, 6 weeks, 3 months, and 6 months. This is temporary and helps increase milk supply.

Common Breastfeeding Challenges

- **Sore nipples:** Most common in the first week. Usually caused by a poor latch. Correct the latch, apply expressed breast milk to nipples after feeding, allow them to air-dry. Lanolin cream can help. Persistent pain beyond 1–2 weeks or cracked/bleeding nipples warrants evaluation.
- **Engorgement:** Breasts become overly full, hard, and painful (usually days 3–5 when milk comes in). Feed frequently, apply warm compresses before feeding, cold compresses after, hand express a small amount if needed to soften the breast for latch.
- **Mastitis:** A breast infection causing localized redness, warmth, swelling, pain, and often fever/flu-like symptoms. Continue breastfeeding (it will not harm the baby). Contact your doctor, as antibiotics may be needed.
- **Low milk supply concerns:** True low supply is rare. The best way to increase supply is to feed more frequently and ensure good latch. Adequate hydration and nutrition for the mother are important. Pumping after feedings can also stimulate additional production.
- **Thrush:** A yeast infection that can affect the baby's mouth (white patches on tongue, inner cheeks) and the mother's nipples (pink, flaky, itchy, burning pain). Both mother and baby need treatment — contact your pediatrician.

Pumping and Storing Breast Milk

- **Room temperature:** Freshly expressed milk is safe for up to 4 hours at room temperature (up to 77°F / 25°C)
- **Refrigerator:** Up to 4 days at 40°F (4°C) or lower. Store in the back, not the door.
- **Freezer:** Up to 6 months is optimal; up to 12 months is acceptable. Use freezer-safe breast milk storage bags.
- **Thawing:** Thaw in the refrigerator overnight, or hold under warm running water. Never microwave breast milk (creates hot spots, destroys nutrients). Use thawed milk within 24 hours. Never refreeze thawed milk.

Source: AAP, [HealthyChildren.org](https://www.healthychildren.org) — Breastfeeding Basics; CDC Breast Milk Storage Guidelines

5. Developmental Milestones by Age

Developmental milestones are behaviors or skills that most children achieve by a certain age. They cover four areas: social/emotional, language/communication, cognitive (learning and problem-solving), and physical/motor development. Every child develops at their own pace, so these are general guidelines, not rigid deadlines.

2 Months

- Social: Begins to smile at people, can briefly calm self (may bring hands to mouth)
- Language: Coos, makes gurgling sounds, turns head toward sounds
- Cognitive: Pays attention to faces, begins to follow objects with eyes
- Motor: Can hold head up during tummy time, makes smoother arm and leg movements

4 Months

- Social: Smiles spontaneously, likes to play with people, may cry when play stops
- Language: Begins to babble, babbles with expression, copies sounds
- Cognitive: Reaches for toys with one hand, follows moving objects with eyes from side to side
- Motor: Holds head steady without support, pushes down on legs when feet are on hard surface, may roll from tummy to back

6 Months

- Social: Knows familiar faces, likes to look at self in mirror, responds to emotions
- Language: Responds to own name, strings vowels together when babbling (ah, eh, oh), responds to sounds by making sounds
- Cognitive: Brings things to mouth, shows curiosity, tries to get things out of reach
- Motor: Rolls in both directions, begins to sit without support, supports weight on legs when standing with help

9 Months

- Social: May be clingy with familiar adults, has favorite toys, understands 'no'
- Language: Understands 'no,' makes many different sounds (mamamama, dadadada), copies sounds and gestures
- Cognitive: Watches the path of falling objects, looks for hidden objects (object permanence), plays peek-a-boo
- Motor: Stands holding on, sits without support, pulls to stand, crawls

12 Months

- Social: Cries when parent leaves, is shy or nervous with strangers, plays simple games like pat-a-cake

- Language: Says 'mama' and 'dada' with meaning, tries to say words you say, responds to simple spoken requests
- Cognitive: Explores objects in different ways (shaking, banging, throwing), finds hidden objects, looks at the correct picture when named
- Motor: May stand alone, may take a few steps without holding on, gets to a sitting position without help

18 Months

- Social: May have temper tantrums, shows affection to familiar people, plays simple pretend
- Language: Says several single words, says and shakes head 'no,' points to show others something interesting
- Cognitive: Knows what ordinary objects are for (phone, brush, spoon), points to get attention, scribbles on own
- Motor: Walks alone, may walk up steps, pulls toys while walking, drinks from a cup, eats with a spoon

2 Years

- Social: Copies others especially adults and older children, gets excited with other children, shows increasing independence
- Language: Points to things in a book, says 2–4 word sentences, follows simple instructions, knows names of familiar people and body parts
- Cognitive: Finds hidden objects, begins to sort shapes and colors, completes sentences in familiar books
- Motor: Stands on tiptoe, kicks a ball, begins to run, walks up and down stairs with support, throws ball overhand

When to Talk to Your Doctor About Development

NOTE: Talk to your pediatrician if your child: does not respond to their name by 12 months, does not point to objects by 14 months, does not walk by 18 months, does not speak at least 6 words by 18 months, does not speak 2-word phrases by 24 months, loses skills they once had at any age, or if you have any concerns. Early intervention makes a significant difference.

Source: CDC Learn the Signs. Act Early. Program; AAP Developmental Surveillance Guidelines

6. When to Go to the Emergency Room

Knowing when to call 911, when to go to the ER, and when to call your pediatrician can save critical time in an emergency. When in doubt, always err on the side of calling.

Call 911 Immediately For:

ALERT: These are life-threatening emergencies. Call 911 first, then begin first aid.

- **Not breathing or severe difficulty breathing** — gasping, grunting, unable to speak or cry
- **Unconscious or unresponsive** — cannot be woken up
- **Choking** with inability to breathe, cough, or cry
- **Seizure lasting more than 5 minutes**, or first-time seizure
- **Severe allergic reaction (anaphylaxis)** — swelling of face/throat, difficulty breathing, hives spreading rapidly. Use EpiPen if available, then call 911.
- **Severe bleeding** that does not stop with 10 minutes of direct pressure
- **Suspected poisoning** — Call Poison Control at **1-800-222-1222** AND call 911 if the child is unconscious, having seizures, or having difficulty breathing
- **Near drowning** (even if the child seems fine afterward)
- **Head injury** with loss of consciousness, confusion, vomiting, unequal pupils, or difficulty walking
- **Blue or gray lips, tongue, or skin**
- **Burns** covering a large area, burns to the face/hands/feet/groin, or chemical/electrical burns
- **Suicidal statements or self-harm** — also call/text **988** (Suicide and Crisis Lifeline)

Go to the ER (Drive or Urgent Care) For:

- Fever in any infant under 3 months (100.4°F / 38°C or higher) — this can also warrant a 911 call if the baby looks ill
- Fever in any child who looks very ill, is lethargic, or is unresponsive
- Severe vomiting or diarrhea with signs of dehydration (sunken eyes, no tears, no wet diapers for 6+ hours)
- Suspected broken bone (visible deformity, inability to move the limb, severe swelling)
- Cuts that are deep, gaping, on the face, or won't stop bleeding — may need stitches
- Animal bites that puncture the skin
- Severe abdominal pain, especially if localized to the right lower side
- Stiff neck with fever and headache
- Spreading purple or red rash with fever (could indicate meningococemia or other serious infection)
- Difficulty breathing (wheezing not responding to asthma inhaler, breathing fast, ribs visible with each breath)
- Blood in stool or vomit
- High fever (above 104°F / 40°C) not responding to medication

- Testicle pain or swelling (possible torsion — time-sensitive)

Call Your Pediatrician (Not Necessarily the ER) For:

- Fever in a child over 3 months who is alert and drinking fluids
- Mild to moderate ear pain
- Rash without fever (non-urgent but worth discussing)
- Vomiting or diarrhea without dehydration signs
- Cold symptoms (runny nose, cough) that are not causing breathing difficulty
- Minor injuries (small cuts, scrapes, bruises)
- Constipation
- Questions about medications or treatments

Important Phone Numbers to Post at Home

- **911** — Life-threatening emergencies
- **1-800-222-1222** — Poison Control Center (free, 24/7, staffed by toxicology experts)
- **988** — Suicide and Crisis Lifeline (call or text, 24/7)
- **Your pediatrician's phone number** — most offices have an after-hours nurse line

Source: AAP, HealthyChildren.org — When to Call EMS; First Aid for Families (PedFACTs)

7. Nutrition by Age

0–6 Months: Breast Milk or Formula Only

- Breast milk is the sole recommended nutrition for the first 6 months. No water, juice, cereal, or other foods are needed.
- Formula-fed infants: Use iron-fortified infant formula. Follow preparation instructions exactly. Do not dilute formula.
- Do NOT give honey, cow's milk, or water to infants under 12 months.
- Vitamin D supplement: 400 IU daily for breastfed and partially breastfed infants.

6–8 Months: Introducing Solid Foods

Signs of readiness: the baby can sit up with minimal support, has good head control, shows interest in food, opens mouth when food is offered, and has lost the tongue-thrust reflex (no longer pushes food out of the mouth with the tongue).

- **Start with single-ingredient purees:** iron-fortified infant cereal, pureed vegetables (sweet potato, squash, peas, green beans), pureed fruits (banana, avocado, pear, applesauce), pureed meats.
- **Introduce one new food every 3–5 days** to watch for allergic reactions (rash, vomiting, diarrhea, swelling).
- **Early allergen introduction:** Current guidelines recommend introducing common allergens (peanut products, eggs, dairy, wheat, soy, tree nuts, fish, shellfish) early, around 6 months, as this may REDUCE the risk of developing food allergies. Peanut-containing foods should be introduced around 6 months for all infants, especially those with severe eczema or egg allergy (consult your pediatrician first for high-risk infants).
- **Texture:** Start with smooth purees, then progress to mashed and soft lumpy textures over the coming months. Let the baby set the pace.
- **Continue breast milk or formula** as the primary nutrition source. Solid foods are supplementary at this stage.

8–10 Months: Expanding Variety and Texture

- Introduce soft finger foods: small pieces of banana, cooked peas, soft cheese, scrambled eggs, small pieces of toast.
- Offer a sippy cup with water at meals (small amounts, not as a replacement for breast milk or formula).
- Encourage self-feeding. Expect messes — this is a crucial part of development.
- Continue to offer a wide variety of flavors and textures. Repeated exposure (10–15 times) may be needed before a baby accepts a new food.

12–24 Months: Transitioning to Table Foods

- **Whole milk:** Switch from formula or breast milk to whole cow's milk at 12 months. Limit to 16–24 ounces per day. Excess milk can displace solid food intake and cause iron deficiency.
- **No honey until 12 months** (risk of infant botulism). After 12 months, honey is safe.

- **Choking hazards to avoid:** whole grapes (cut lengthwise into quarters), hot dogs (cut lengthwise then into small pieces), popcorn, whole nuts, hard candy, large chunks of raw vegetables or hard fruits, spoonfuls of peanut butter (spread thinly instead).
- **Daily diet should include:** grains, fruits, vegetables, protein (meat, beans, eggs), dairy. Offer small portions and let the child decide how much to eat.
- **Juice:** Not recommended before 12 months. After 12 months, limit to 4 ounces per day of 100% fruit juice. Whole fruits are always preferred over juice.
- **Avoid added sugars and excessive salt.** No sugar-sweetened beverages at any age.

2–5 Years: Establishing Healthy Patterns

- Serve 3 meals and 2–3 healthy snacks per day at regular times.
- Offer a variety of colors and food groups. Children often prefer plain, simple foods.
- **Picky eating is normal.** Continue offering rejected foods without pressure. It can take 10–15 exposures for a child to accept a new food. Do not force, bribe, or use food as reward.
- **Milk:** Switch to low-fat (1%) or fat-free milk at age 2. Limit to 16–20 ounces per day.
- **Portion sizes:** A child's serving is about 1 tablespoon per year of age for each food. Start small — the child can ask for more.
- **Limit screen time during meals.** Eat together as a family when possible. Children who eat family meals tend to have healthier diets.

6–12 Years: Growing Independence

- Children need about 1,200–2,000 calories per day depending on age, sex, and activity level.
- Ensure adequate calcium (3 servings of dairy per day or calcium-fortified alternatives) for bone growth.
- Iron-rich foods are important, especially for girls approaching puberty: lean meats, beans, fortified cereals, dark leafy greens.
- Encourage water as the primary beverage. Avoid soda and energy drinks entirely.
- Teach children to read nutrition labels and make healthier choices.

13–18 Years: Teen Nutrition

- Calorie needs increase significantly: 1,800–3,200 calories per day depending on sex, size, and activity.
- Calcium needs peak at 1,300 mg per day (critical for bone density).
- Iron needs increase, especially for menstruating teens (15 mg per day for girls vs 11 mg for boys).
- Watch for warning signs of eating disorders: rapid weight loss, obsession with food or body image, avoiding meals, excessive exercise, or purging behaviors. Contact your pediatrician if concerned.
- Energy drinks are not recommended for any children or adolescents. They contain excessive caffeine and stimulants that can cause heart palpitations, anxiety, and sleep disruption.

Source: AAP Nutrition Guidelines; HealthyChildren.org — Nutrition by Age; CDC Nutrition Resources

Disclaimer: This document is compiled for educational and informational purposes as part of the PediatricAI project. It is based on publicly available guidelines from the AAP, CDC, and WHO. It is NOT a substitute for professional medical advice, diagnosis, or treatment. Always seek the advice of your child's pediatrician with any questions regarding a medical condition.